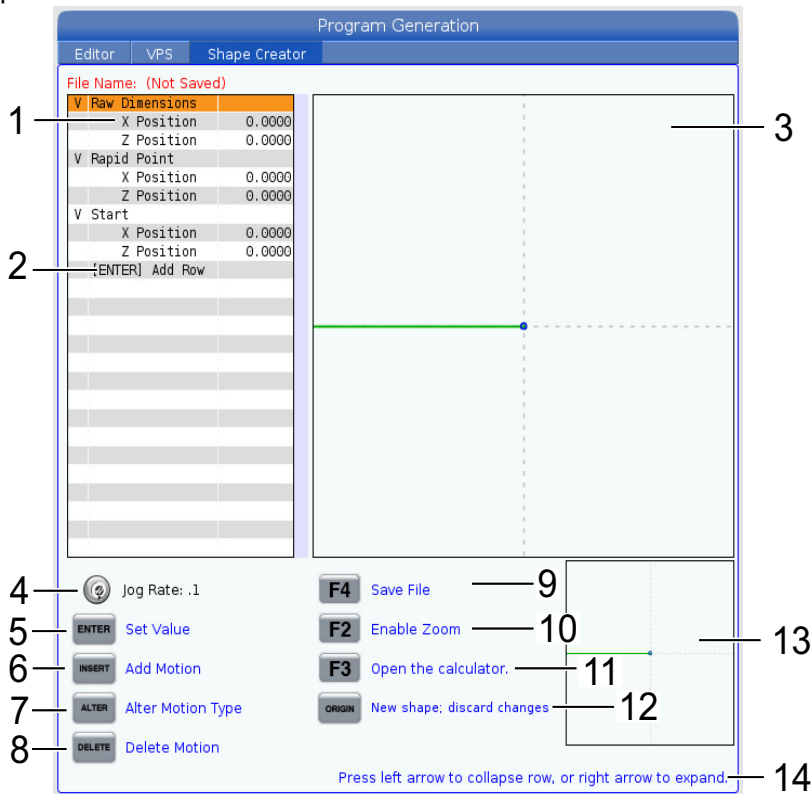


6.8 Shape Creator

Shape creator lets you quickly draw program shapes and tool paths. To create a new shape, press **[EDIT]** and then select the **Shape Creator** tab. If you already created a shape profile, go to list program User Data, My Profiles folder and select the shape creator file. Press **[SELECT PROGRAM]** to continue editing the shape.

F6.11: Shape Creator Screen.



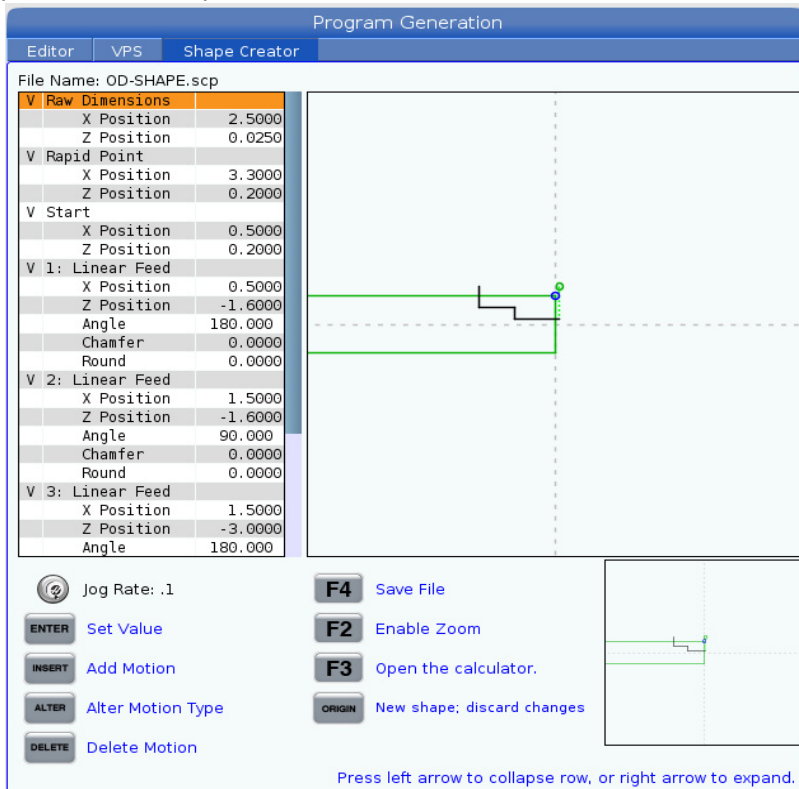
1. Motion variables.
2. Press **[ENTER]** to add a new row.
3. Shape Creator drawing board.
4. Handle Jog Rate
5. Press **[ENTER]** to set the value.
6. Press **[INSERT]** to insert a motion: Linear Feed Motion, CW Circular Feed Motion, CCW Circular Feed Motion.
7. Highlight the desired Motion and Press **[ALTER]** button to change to a different motion type.
8. Highlight the desired Motion and Press **[DELETE]** button to delete the motion.

9. Press **[F4]** then type in a name to save the Shape Generator file. It will save the file in the **User Data/My Profiles/** folder.
10. Press **[F2]** button to Enable Zoom
11. Press **[F3]** button to open the calculator function.
12. Press **[ORIGIN]** button to do a new shape or discard the changes made.
13. Zoom view box.
14. Help text.

6.8.1 Using Shape Creator

The following is an example using shape creator to generate a simple OD rough turning profile.

F6.12: Shape Creator Example.



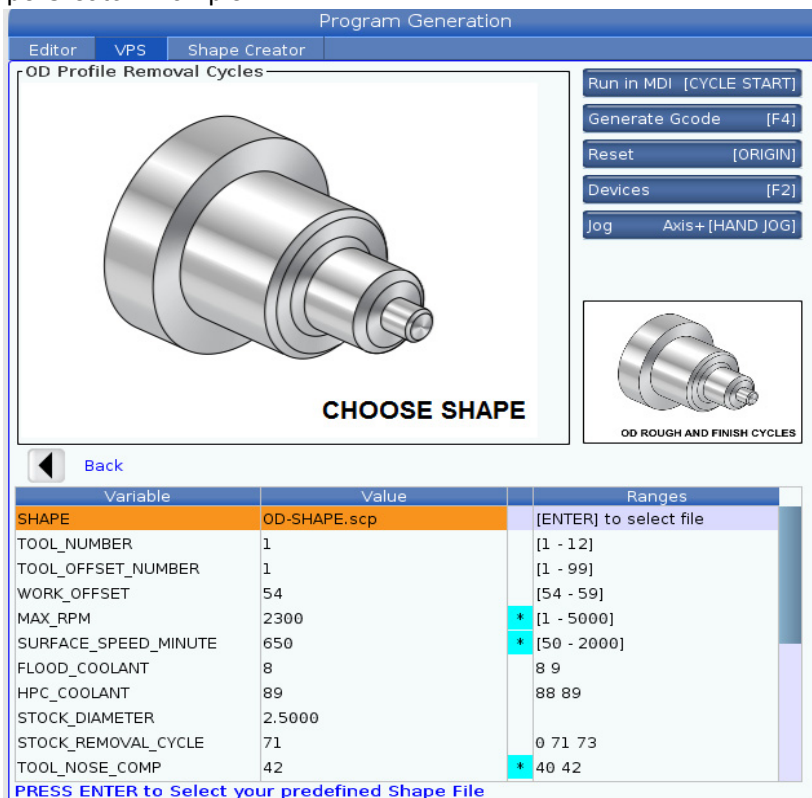
1. Press the **[EDIT]** button and right cursor to the Shape Creatortab.
2. Use the jog wheel to set the values. Set the Raw Dimension Positions: **X Position 2.5000, Z Position 0.0250**.
3. Set the Rapid Point Positions: **X Position 3.3000, Z Position 0.2000**.
4. Set the Start Positions: **X Position 0.5000, Z Position 0.2000**.

5. Press **[ENTER]** to Add a row and select **1: Linear Feed Motion**.
6. Set the Linear Feed Positions: **X Position** 0.5000, **Z Position** -1.6000, **Angle**180.000, **Chamfer** 0.0000, **Round** 0.0000.
7. Press **[ENTER]** to Add a row and select **1: Linear Feed Motion**.
8. Set the Linear Feed Positions: **X Position** 1.5000, **Z Position** -1.6000, **Angle**90.000, **Chamfer** 0.0000, **Round** 0.0000.
9. Press **[ENTER]** to Add a row and select **1: Linear Feed Motion**.
10. Set the Linear Feed Positions: **X Position** 1.5000, **Z Position** -3.0000, **Angle**180.000, **Chamfer** 0.0000, **Round** 0.0000.
11. Press **[ENTER]** to Add a row and select **1: Linear Feed Motion**.
12. Set the Linear Feed Positions: **X Position** 3.3000, **Z Position** -3.0000, **Angle**90.000, **Chamfer** 0.0000, **Round** 0.0000.
13. Press **[F4]** to save the shape profile. When finished the control will save the file to the User Data Tab, My Profiles Folder. See the next section to generate the G-Code program using the VPS template using this shape profile.

6.8.2 Using Shape Creator - VPS Template

This example will create a G-code program using the OD Profile Removal Cycles in the VPS templates.

F6.13: Shape Creator Example.



1. Press the **[EDIT]** button and left cursor to the VPS tab.
2. Go to the VPS folder and press the right cursor to see the templates.
3. Find the OD Profile Removal Cycles template and press the **[ENTER]** button.
4. For **SHAPE** press **[ENTER]** to select the Shape Creator file that was created on the previous section.
5. Set the **TOOL_NUMBER** to 1.
6. Set the **TOOL_OFFSET_NUMBER** to 1.
7. Enter the **WORK_OFFSET** number. In this example the value is **54**.
8. Set the **MAX_RPM** variable to: **2300**
9. Set the **SURFACE_SPEED_MINUTE** variable to: **650**
10. Set the **FLOOD_COOLANT** to: **8**.
11. Set the **HPC_COOLANT** variable to: **88**

12. Set the **STOCK_DIAMETER** is defined in the Shape Creator file.
13. Set the **STOCK_REMOVAL_CYCLE** variable to: **71**.
14. Set the **TOOL_NOSE_COMP** (Tool Nose Compensation) variable to: **42**.
15. Set the **DOC** (Depth of Cut) variable to: **0.05**
16. Set the **X_FINISH_STOCK** variable to: **0.01**
17. Set the **Z_FINISH_STOCK** variable to: **0.003**
18. Set the **FEEDRATE** variable to: **0.01**
19. The **X_RAPID_POINT** is defined in the Shape Creator file.
20. The **Z_RAPID_POINT** is defined in the Shape Creator file.
21. For the **RETRACT_X_HOME** enter **Y** to send the turret home in the X-Axis or **N** to enter the Z-Axis Tool Change position value on the next line.
22. For the **RETRACT_Z_HOME** enter **Y** to send the turret home in the Z-Axis or **N** to enter the Z-Axis Tool Change position value on the next line.
23. Set the **END_M_CODE** variable to: **30** to end the program with an M30.
24. Press **[F4]** to Generate the G-code, and select **2** to Output to MDI.
25. Press the **[GRAPHICS]** button. Run the program and verify that the program runs with no alarms.